



Detecting sexism in social media: an empirical analysis of linguistic patterns and strategies

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Abstract

With the rise of social networks, there has been a marked increase in offensive content targeting women, ranging from overt acts of hatred to subtler, often overlooked forms of sexism. The EXIST (sEXism Identification in Social neTworks) competition, initiated in 2021, aimed to advance research in automatically identifying these forms of online sexism. However, the results revealed the multifaceted nature of sexism and emphasized the need for robust systems to detect and classify such content. In this study, we provide an extensive analysis of sexism, highlighting the characteristics and diverse manifestations of sexism across multiple languages on social networks. To achieve this objective, we conducted a detailed analysis of the EXIST dataset to evaluate its capacity to represent various types of sexism. Moreover, we analyzed the systems submitted to the EXIST competition to identify the most effective methodologies and resources for the automated detection of sexism. We employed statistical methods to discern textual patterns related to different categories of sexism, such as stereotyping, misogyny, and sexual violence. Additionally, we investigated linguistic variations in categories of sexism across different languages and platforms. Our results suggest that the EXIST dataset covers a broad spectrum of sexist expressions, from the explicit to the subtle. We observe significant differences in the portrayal of sexism across languages; English texts predominantly feature sexual connotations, whereas Spanish texts tend to reflect neosexism. Across both languages, objectification and misogyny prove to be the most challenging to detect, which is attributable to the varied vocabulary associated with these forms of sexism. Additionally, we demonstrate that models trained on platforms like Twitter can effectively identify sexist content on less-regulated platforms such as Gab. Building on these insights, we introduce a transformer-based system with data augmentation techniques that outperforms competition benchmarks. Our work contributes to the field by enhancing the understanding of online sexism and advancing the technological capabilities for its detection.

Keywords Online sexism · Sexism categorization · Social media · Sexism dataset

1 Introduction

The Oxford English Dictionary defines sexism as “prejudice, stereotyping or discrimination, typically against women, on the basis of sex”. Despite substantial progress in gender

equality, including the narrowing of the wage gap, the increased representation of women in leadership roles, and the implementation of educational programs to prevent domestic violence, inequality and discrimination against women persist in society [1, 2]. While violent and hateful forms of sexism are widely recognized and condemned, several studies point to a progressive normalization of some forms of “everyday” sexism, especially among adolescents [3].

The concept of “ambivalent sexism”, introduced by Peter Glick and Susan Fiske in 1996, captures two forms of sexism: hostile and benevolent. Hostile sexism is overtly negative and involves hostility and aggression toward women. In contrast, benevolent sexism takes on a more positive tone but still implies that women are inferior to men due to their vulnerability or need for assistance and protection. Although

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society generally rejects hostile sexism, benevolent sexism is frequently accepted and normalized.

Subtle and benevolent forms of sexism continue to persist in social networks, often shielded by anonymity. Given the considerable number of teenagers online, it is crucial to promptly investigate and address the growing issue of online sexism. Natural language processing (NLP) systems present a potential solution for identifying subtle expressions of sexism and raising awareness about them. However, for these systems to be effective, it is important to understand how sexism manifests in social networks, determine which users are most affected by online sexism, identify what aspects of women are being targeted, and evaluate the scope and impact of these messages.

In 2021, the sEXism Identification in Social neTworks (EXIST) shared task was proposed at the IberLEF 2021 forum [4]. The EXIST challenge was the first to address the identification and categorization of sexism in the broadest sense, from overtly hostile to subtle and even benevolent expressions. Earlier research on online sexism primarily targeted the detection of misogyny and woman-directed hatred but disregarded subtle sexism [5–7].

The competition provided a benchmark dataset, allowing the community to evaluate various methods for sexism identification and categorization. This dataset, consisting of more than 11,000 tweets and gabs (messages from Gab.com) in English and Spanish, contains sexist and non-sexist expressions and categorizes the sexist messages according to the aspects of the women that they attack. Thirty-one research teams from 11 countries presented a variety of algorithms and techniques to address the two different subtasks: sexism identification and sexism categorization. The outcomes and challenges highlighted by the participants show the complexity of detecting sexism, indicating a need for further research and more reliable systems.

Building on the EXIST 2021 framework, EXIST 2022 [8] employed the same approach and tasks as its predecessor of 2021, incorporating a new test set of 1,058 tweets. The EXIST 2021 dataset was used for the training set, and this new dataset of 1,058 tweets was used as the test set. In this study, we concentrate on the EXIST 2021 competition because the test dataset is larger, providing more stable results and conclusions. Using this dataset allows us to draw equivalent inferences for the EXIST 2022 competition, as in this competition, most of the data was included in EXIST 2021.

In this article, we aim to analyze the data gathered for the competition, as well as the systems and results presented by the participants, to answer the following research questions:

1. Research question 1: Does the EXIST dataset properly capture both hateful and benevolent expressions of sexism? To answer this question, we analyze the presence of

hateful terms in the dataset and use a hate-speech classifier to detect offensive texts. Our hypothesis is that the dataset contains a significant number of sexist expressions that do not use any hateful words and that the hate-speech classifier will not be able to adequately capture all the sexist expressions, thus demonstrating the presence of subtle, nonhateful sexism in the dataset.

2. Research question 2: Which algorithms, techniques, and resources are more suitable for detecting sexist expressions in social networks? To answer this question, an exhaustive comparison of all the systems employed by the participants is performed. The techniques are divided into different dimensions and discussed separately.
3. Research question 3: Are all categories of sexism equally difficult to predict automatically? To answer this question, we assess the systems' capacity to detect different types of sexism. In addition, we study the vocabulary homogeneity within each category to explain the differences in performance.
4. Research question 4: Are the characteristics of sexist messages consistent among languages, or does language influence the capacity of automatic systems to detect sexism? To answer this question, we evaluate the ability of the systems to detect sexism in English and Spanish separately. Furthermore, we study the differences between languages by studying the data distribution and vocabulary homogeneity.
5. Research question 5: Does the detection of sexism depend on the social network where the message is posted? To answer this question, we test the capacity of the systems proposed in the competition to detect sexism in Gab after training on messages from a different social network, Twitter.
6. Research question 6: Are the findings from the EXIST competition helpful in predicting sexism online? To answer this question, we apply the lessons learned from the analysis of the EXIST data to the development of a new sexism detection system.

The main contributions of this work are, therefore, the following findings, which are expected to guide the future development of sexism detection systems and the advancement of the discipline:

1. Our comprehensive analysis of the EXIST dataset shows that the EXIST dataset contains various forms of sexism, expressed in both explicit and implicit ways. We discovered that some of the most harmful types of sexism, such as misogyny and sexual violence, can be communicated without the use of explicitly offensive language. This renders automatic systems that rely solely on the detection of hate speech inadequate for detecting sexism.

2. We confirmed that the most effective approaches for detecting sexism in text utilize transformer-based systems in conjunction with data augmentation techniques in monolingual settings. The main challenge that these systems encounter is the lack of contextual information and the inherent subjectivity of the task, which can be addressed by employing data augmentation techniques.
3. Our analysis indicates that different forms of sexism are expressed through different linguistic patterns. Specifically, misogyny is characterized by a broader and more diverse vocabulary, whereas other forms, such as sexual violence and ideological sexism, tend to employ more consistent and uniform terminology. This distinction suggests that certain types of sexism may be more readily identifiable due to their more predictable language use.
4. Our analysis of the variations in expressing sexist attitudes across different languages reveals that Spanish texts tend to exhibit a slightly more homogeneous vocabulary. However, there are notable differences in the prevalence of various sexism categories between English and Spanish. In Spanish, contemporary forms of sexism and attacks on feminism appear to be more prominent, whereas English texts predominantly feature sexual connotations. Despite these differences, the system's effectiveness in detecting such subtypes of sexism remains consistent across languages. This observation suggests that the most challenging sexist subtypes to detect may be independent of language.
5. Despite differences in the hateful content policy and user characteristics between Twitter and Gab, our findings indicate that it is possible to train the systems successfully on one network (i.e., Twitter) and use them to detect sexism in another network (i.e., Gab). This demonstrates the feasibility of cross-platform training between Twitter and Gab and indicates the robustness of these systems to variations in the online environment.
6. Based on these findings, we developed a transformer-based system composed of two different modules for English and Spanish. This system, pre-trained on unlabeled tweets and fine-tuned with the EXIST dataset, sets new benchmarks for sexism detection, showcasing our technical innovation and contribution to advancing the discipline.

The remainder of this paper is organized as follows: in Section 2, we discuss related works. In Section 3, we describe the experiments performed to answer our different research questions and analyze the results. In Section 4, the results for the newly proposed classification system are presented and analyzed. Finally, the conclusions and future work are presented in Section 5.

2 Related work

The automatic detection of sexism in online conversations is receiving increasing attention from the scientific community. However, most of the related research has focused on misogyny detection [9–12], while few studies have addressed sexism in its broadest sense. In fact, detection of hate speech against women and misogyny are often confused with sexism detection [13]. Misogyny can be considered as a subtype of hate speech and is defined as “hateful content targeting women” [9]. Although the misogyny and sexism detection tasks are closely related and both target women, they have different nuances. While misogyny involves expressions of hostility and hatred toward women, sexism includes any form of oppression or prejudice against women and can therefore be hostile (as in the case of misogyny) or subtle.

One of the earliest works to categorize misogyny in social media was [12]. In this work, misogyny is classified into five subcategories: discredit, harassment/threats of violence, derailing, stereotype/objectification, and dominance. Another relevant study is the AMI shared task [5], which aimed to detect misogyny automatically on Twitter in both Spanish and English. The methods presented in the competition were primarily based on supervised machine learning using different textual features (such as unigrams and bigrams, sentiment-based information, or syntactic categories) and user-based features (such as the number of retweets, followers, etc.), along with traditional machine learning algorithms such as logistic regression, support vector machines, and decision trees [18–20].

Recently, new taxonomies and datasets for misogyny detection have been developed. Guest et al. [7] constructed an expert-labeled dataset composed of 6,567 messages extracted from Reddit posts and comments. Additionally, they proposed a new taxonomy for misogyny labeling inspired by [12], which comprises four different categories: misogynistic pejoratives, descriptions of misogynistic treatment, acts of misogynistic derogation, and gendered personal attacks against women. Similarly, [14] proposed a new set of labels for annotating misogyny in natural written language and a new dataset. Their fine-grained categorization of misogyny differentiated the target group and other abusive phenomena such as racism or homophobia. They proposed six categories: neosexism, discredit, stereotypes and objectification, benevolent sexism, dominance, and sexual harassment.

Because misogyny is a form of hatred directed toward women, previous studies on misogyny identification have employed techniques and methodologies from the field of automated hate speech detection. In many cases, misogyny has been analyzed in conjunction with other types of hate speech, as in SemEval 2019 (Task 5) [15], the aim of which was to detect hate speech against immigrants and women in

Spanish and English messages retrieved from Twitter. This task included a corpus of 19,600 tweets, 13,000 in English and 6,600 in Spanish, that were annotated by crowdsourced workers. Deep learning methods were presented to the competition for identifying racism and misogyny in texts [21].

Pioneering studies on the detection of sexism also employed techniques and resources taken from the hate speech research field [9, 10]. One of the first datasets of sexist expressions was developed for studying sexism in conjunction with racism [10]. However, again this dataset only comprises expressions of hatred or hostility toward women, overlooking subtle sexism. Professional annotators identified sexist and racist tweets according to critical race theory [22]. The tweets were initially labeled sexist, racist, or neither by the authors of the study and then reviewed by a third annotator. The resulting dataset consists of 17k tweets, of which only 20% are labeled as sexist. The work of [9] extended the dataset proposed by [10] with 4,033 new tweets and showed the impact of allowing majority voting on classification and agreement between the amateur and expert annotators. As in other domains [23, 24], they proposed a machine learning classification system based on bag-of-words along with traditional machine learning algorithms such as logistic regression.

Rodríguez-Sánchez et al. [13] released the first Spanish corpus of sexist expressions on Twitter, the MeTwo dataset that considers subtle sexism. The dataset, composed of 3600 tweets, was intended only for sexism detection and was labeled based on a majority vote by three annotators. In addition, they compared traditional and neural network-based methods and showed that BERT outperformed the other algorithms tested. Jiang et al. [16] proposed the first Chinese dataset to detect sexism online (the SWSR dataset). In particular, given the limited presence of Chinese content on Twitter and geographical access restrictions regarding it, the authors focused on the most prevalent microblogging platform in China, Sina Weibo. They also proposed a new sexism-related

offensive lexicon, SexHateLex. More recently, [17] developed CoRoSeOf, a large corpus of Romanian social media messages manually annotated following a two-level labeling scheme. First, they divided content into two categories: “sexist” and “non-sexist”. Second, they used three categories for sexist messages, “sexist direct”, “sexist descriptive”, and “sexist reporting”, and two categories for non-sexist posts, “non-offensive” and “offensive”. The final dataset contains 39,245 tweets, annotated by a team of ten annotators (seven women and three men).

For clarity and to facilitate comparison, Table 1 summarizes the most relevant works reviewed in this section. The present work takes advantage of the dataset built for the shared task EXIST 2021 at IberLEF 2021 [4]. The dataset was designed to identify any kind of sexism in a multilingual setting and is composed of more than 11,000 texts. To the best of our knowledge, no previous work has explicitly addressed the analysis of sexism characteristics, from subtle to hostile expressions, and their subtypes on social networks.

3 Experimental methodology

In this section, we describe various experiments for investigating online sexism using a dataset from the EXIST 2021 competition. First, Section 3.1 introduces the EXIST 2021 tasks and data. Section 3.2 addresses the main research questions presented in the introduction. In Section 3.2.1, we evaluate the coverage and effectiveness of the EXIST dataset in capturing both overt and nuanced forms of sexism. Sections 3.2.2, 3.2.3, and 3.2.4 investigate the linguistic characteristics of the different subtypes of sexism and their impact on performance in cross-lingual and cross-platform settings. Section 3.2.5 then examines the different classification methods presented in the EXIST competition. Finally, building on all the insights from this section, Section 3.3

Table 1 Research papers related to hate speech or sexism detection

Paper	Domain	Language	Output
Waseem [9]	Twitter	English	Hate speech: women, racism
Waseem and Hovy [10]	Twitter	English	Hate speech: women, racism
Frenda et al. [11]	Twitter	English	Misogyny
Anzovino et al. [12]	Twitter	English and Spanish	Misogyny
Rodríguez-Sánchez et al. [13]	Twitter	Spanish	Sexism
Guest et al. [7]	Reddit	English	Misogyny
Zeinert et al. [14]	Twitter, Facebook and Reddit	Danish	Misogyny
Basile et al. [15]	Twitter	English and Spanish	Hate speech: immigrants, women
Jiang et al. [16]	Sina Weibo	Chinese	Sexism
Hofels et al. [17]	Twitter	Romanian	Sexism
Rodríguez-Sánchez et al. [4]	Twitter and Gab	English and Spanish	Sexism

introduces a language-specific, transformer-based detection system to leverage unlabeled in-domain data.

3.1 EXIST 2021: task description and dataset

The EXIST 2021 shared task is the most suitable framework for investigating our research questions. First, the dataset enables us to analyze how sexism is expressed across various languages and platforms. Second, it contains instances of both hostile and subtle sexism. Third, the shared task provides us with an opportunity to identify the challenges that existing systems face in detecting and classifying sexism.

The EXIST 2021 competition is defined by the organizers [4] as a multilingual classification task that includes two main subtasks: (i) sexism identification (task 1) and (ii) sexism categorization (task 2).

Task 1 is defined as a binary classification problem, where automatic systems must determine whether a message is sexist according to the following definitions:

- **Sexist:** the message expresses sexist behaviors or discourses, including the report or description of sexist events, for example, *“People really try to convince women with little to no ass that they should go out and buy a body. Like bjh, I don’t need a fat ass to get a man.”*.
- **Non-Sexist:** the text does not express any sexist behavior or discourse, for example, *“it’s honestly so embarrassing to watch and they’ll be like “not all white women are like that””*.

Task 2 of EXIST aims to categorize a message that has been classified as sexist according to the type of sexism it conveys. The categorization system was reviewed by two experts in gender issues, Trinidad Donoso and Miriam Comet from the University of Barcelona, who considered the various ways in which women are undermined. This task involves a multiclass classification problem where each sexist tweet or gab must be assigned to one of the following five categories:

- **Ideological and inequality:** The text discredits the feminist movement, rejects the idea that there is inequality between men and women or presents men as victims of gender-based oppression, for example, *“#Feminism is a war on men, but it’s also a war on women. It’s a war on female nature, a war on femininity”*. In this study, we adhere to the definition of feminism as “belief in and advocacy of the political, economic, and social equality of the sexes”.¹ Therefore, texts that discredit or attack feminism undermine this pursuit of equality and are thus considered sexist.

¹ <https://www.merriam-webster.com/dictionary/feminism>

- **Stereotyping and dominance:** The text expresses false ideas about women that suggest that they are more suitable for fulfilling certain roles (mother, wife, family caregiver, faithful, tender, loving, submissive, etc.) or are not suited for certain tasks (driving, hard work, etc.) or claim that men are somehow superior to women, for example, *“Woman driving, be careful!”*.
- **Objectification:** The text presents women as objects apart from their dignity and individual aspects or assumes or describes certain physical qualities that women must have in order to fulfill traditional gender roles (compliance with beauty standards, hypersexualization of female attributes, women’s bodies being at the disposal of men, etc.), for example, *“No offense but I’ve never seen an attractive african american hooker. Not a single one”*.
- **Sexual violence:** Sexual suggestions, requests for sexual favors, or harassment of a sexual nature (rape or sexual assault) are made, for example, *“I wanna touch your tits..you can’t imagine what I can do on your body”*.
- **Misogyny and non-sexual violence:** The text expresses hatred and violence toward women. For example *“I hate misogyny more than I hate women”*.

The EXIST dataset comprises data from Twitter and Gab.com in English and Spanish. While Twitter data were used for both training and testing, Gab data were only included in the EXIST test set to analyze the differences between social networks with and without content control mechanisms. To provide training and testing data, experts collected and analyzed terms commonly used to undermine the role of women in our society, which were subsequently filtered. Crowdworkers annotated the tweets using Amazon Mechanical Turk, with each tweet/gab being assessed by five annotators. The final label was assigned using a majority vote.

The EXIST dataset is, to the best of our knowledge, the first multilingual corpus designed to identify sexism in a broad sense, covering hostile, subtle, and benevolent sexism. Table 2 shows some statistics of the dataset, including the number of texts per class for both the training and test sets, as well as the distribution by language. For a more detailed description, please refer to the overview of the EXIST task [4].

3.2 Analysis of the data and systems from the EXIST competition

3.2.1 Assessing the coverage of hateful and benevolent expressions of sexism in the EXIST dataset

As previously mentioned, the EXIST dataset was created to aid in the detection of sexism in online conversations, ranging from explicit hate or violence toward women to subtle

Table 2 EXIST 2021 dataset distribution of instances per classes, languages, and sources in the training and test sets

Category	Training		Test			
	Spanish	English	Spanish Twitter	Spanish Gab	English Twitter	English Gab
Sexist	1741	1636	858	265	858	300
Non-sexist	1800	1800	812	225	858	192
Sexist subcategory						
Ideological-inequality	480	386	215	73	233	100
Misogyny-non-sexual-violence	401	284	199	58	152	63
Objectification	244	256	124	50	121	29
Sexual-violence	173	344	131	71	150	48
Stereotyping-dominance	443	366	189	13	202	60

attacking expressions that can go unnoticed. Therefore, we aim to quantify and characterize the presence of both hateful and subtle expressions in the EXIST dataset to ensure that it is a suitable resource for training and testing systems intended to detect sexism with all its nuances.

We must therefore find both hateful and nonaggressive expressions of sexism in the EXIST dataset. To this end, the HurtLex lexicon of offensive, aggressive, and hateful terms [25] is used. Our hypothesis is that the dataset includes a significant number of sexist expressions that do not contain any words from the lexicon.

The HurtLex lexicon, developed from the Italian lexicon “Le Parole per Ferire” by Tullio de Mauro, contains entries divided into 17 different types of offenses grouped into two categories: conservative (literally offensive words) and inclusive (words that can be used with negative connotations). To identify sexist expressions in the EXIST dataset, we used both the Spanish and English versions of the HurtLex lexicon and filtered for the conservative category. We selected topics related to sexism, including PA (professions and occupations), DMC (moral and behavioral defects), ASM (male genitalia), ASF (female genitalia), PR (prostitution-related words), QAS (words with potentially negative connotations), and CDS (derogatory words). The initial lexicon count resulted in 891 terms for Spanish and 1671 for English. To harmonize the number of terms across the two languages, we employed the Google Cloud API² to translate the English lexicon into Spanish. By integrating only the unique terms from this translation into the existing Spanish lexicon, the count increased to 1778 terms, while the English lexicon remained consistent at 1671 terms.

We created a new feature for each tweet that captures the presence of offensive words from HurtLex. In Spanish, we found 421 out of 1778 terms in the lexicon, while in English, we found 354 out of 1671 terms. Table 3 displays the percentage of sexist and non-sexist tweets that contain at least one word from the lexicon in the training set. The fact that

only 54% of sexist tweets contain offensive terms suggests that the remaining 46% do not include any hateful or aggressive words. This indicates that sexism can also be expressed in a subtle and nuanced manner that may be harder to detect, highlighting the importance of developing systems that can identify all forms of sexism in online conversations.

The results by language are shown in Table 4. The proportion of tweets containing offensive terms is greater in the English subset (60%) than in the Spanish subset (48%), likely due to the greater proportion of messages from the sexual violence category in the English subset, which are usually expressed using foul and coarse vocabulary (see Table 2).

Regarding the various categories of sexism, Table 5 presents the percentage of tweets containing offensive or hateful words for each category. It is evident that categories such as misogyny, objectification, and sexual violence have a greater percentage of offensive terms. On the other hand, the ideological and stereotyping categories have a lower proportion of offensive terms. Table 6 shows the results by language.

Although the HurtLex lexicon can identify hateful and offensive terms, it does not consider the context in which these terms are used. Moreover, even explicit sexism can be expressed without the use of any hateful or offensive terms; for example, the sentence “It’s challenging to find a smart woman” is openly sexist and offensive, yet contains no hateful terms. To address this issue, we designed a second experiment utilizing two distinct classifiers. The first classifier is a traditional machine learning-based algorithm relying on logistic regression using the TF-IDF features proposed by [26]. The second classifier [27] is a cutting-edge model based on transformers specifically designed to detect toxic and hate speech. We hypothesize that these classifiers may struggle to identify all sexist messages due to the presence of nuanced expressions of sexism that elude detection by hate speech or toxicity classifiers.

Table 7 displays the predictions made by both classifiers regarding the English sexist tweets. As shown, when applying the classifier from [26], only approximately 30% of sexist

² <https://cloud.google.com/translate/docs/reference/rest/>

Table 3 Percentage of sexist and not sexist texts containing terms from the Hurltex lexicon

Category	% Texts	Avg. tokens in Hurltex	Avg. number of words	Avg. characters
Non-sexist	45.94	1.5	33.06	194.9
Sexist	54.39	1.6	30.3	179.9

tweets are identified as offensive or hateful. For the most advanced hate speech classifier [27], we observe that almost 50% of the sexist tweets are not detected. This indicates that subtle forms of sexism may not be identifiable by state-of-the-art machine learning models designed to recognize offensive language and hate speech.

To extend this analysis to the Spanish subset, we translated all Spanish tweets into English and repeated the classification process. The results, presented in Table 8, aligned with the findings for the English dataset, with only approximately 20% of sexist tweets being identified as offensive or hateful when using the classifier from [26]. The classifier from [27] detected only 41% of the sexist tweets. Consistent with the experiment involving the HurlLex lexicon, the classification system recognized a greater proportion of hateful or offensive messages in the English dataset than in the Spanish dataset.

The results of the two previous experiments suggest that the EXIST dataset contains both hateful and subtle sexist texts, as a considerable proportion of sexist tweets do not contain any offensive terms from the hate lexicon or are not classified as offensive/hateful by a state-of-the-art machine learning model. These findings highlight the complexity of detecting sexism in social media, as it can manifest in various forms and expressions. However, the dataset remains useful for training models to detect sexism, as it provides a diverse range of sexist expressions that can enhance the accuracy and robustness of these models.

3.2.2 Assessing the predictive difficulty of categories of sexism

Our next experiments aim to evaluate the capacity of automatic systems to detect various forms of sexism, as well as to understand the textual attributes of the different types of sexism (stereotyping, sexual, misogyny, etc.) and the factors that make them challenging to predict.

Table 4 Percentage of sexist and not sexist texts per language containing terms from the Hurltex lexicon

Language	Category	% Texts	Avg. tokens in Hurltex	Avg. number of words	Avg. characters
English	Non-sexist	47.16	1.69	33.2	196.8
	Sexist	60.14	1.8	31.5	185.5
Spanish	Non-sexist	44.72	1.43	32.8	192.9
	Sexist	48.99	1.43	29.1	173.61

Table 9 displays the results of the top 10 systems for task 1, showcasing the F1 scores for both the sexist and non-sexist classes, along with accuracy and macro-F1. Notably, we observe that all 10 systems have greater accuracy in predicting the sexist class than in predicting the non-sexist class. In the case of the sexist class, we see a variation of approximately 2.1% between the best- and worst-performing solutions (with the best result in bold and the worst result underlined). For the non-sexist class, we see a difference of 2.3% between the top- and bottom-performing teams.

Table 10 offers a more insightful comparison, as it reports the F1 score for each category of sexism in task 2. The results demonstrate a significant variation in performance per class across the different teams. However, there is consistency in the ranking of the different classes in terms of performance.

Our findings indicate that the *ideological-inequality* class is the easiest to predict, followed by the *sexual-violence* class, across all 10 systems. The superior performance on the ideological-inequality class can be attributed to the fact that it is the majority category and also uses a relatively homogeneous vocabulary related to the feminist movement and women's rights advocacy. This second reason is also applicable to the *sexual-violence* class, on which the methods show good performance due to the use of coarse and offensive vocabulary.

The *misogyny-non-sexual-violence* class appears to be the most difficult to predict, followed by the *objectification* class. It is noteworthy that one might expect better performance in predicting misogynistic expressions, as misogyny typically entails overt hatred or prejudice against women.

We hypothesize that the unexpected results for this class may be due to the heterogeneous nature of the vocabulary used in the messages. To test this hypothesis, we calculated the Jaccard similarity coefficient [28] for each category of sexism and language, which enabled us to measure the homogeneity of the vocabulary used in the messages with respect to the type of sexism they convey.

Table 5 Percentage of sexist texts per category containing terms from the Hurltex lexicon

Category	% Texts	Avg. tokens in Hurltex	Avg. number of words	Avg. characters
Ideological-inequality	43.99	1.5	37.9	232.6
Misogyny-non-sexual-violence	61.6	1.7	24.7	146.6
Objectification	63.8	1.5	27.2	156.7
Sexual-violence	62.28	1.9	30.1	178.8
Stereotyping-dominance	48.57	1.5	31.9	184.5

Table 11 displays the Jaccard similarity coefficients computed for each category of sexism. The coefficients were calculated separately for each language and then averaged. The *misogyny-non-sexual-violence* category has the lowest similarity coefficient, which suggests a low degree of vocabulary homogeneity (even though it contains many offensive terms). Furthermore, this experiment suggests that the EXIST dataset can capture different instances of misogynistic messages, including both direct or hateful expressions and more subtle forms of hate toward women. This is less evident for the *sexual-violence* category; the improved performance in this category indicates that such text is easier to detect, as it usually uses more overtly hateful language (as is confirmed in Section 3.2.1).

We also performed a manual analysis of the errors made by the best system in the competition. The results of this analysis will help us to understand the characteristics that make sexism and its subtypes difficult to identify. Figure 1 presents the confusion matrix corresponding to the top-performing system from the EXIST 2021 competition, that of the AI-UPV team (as detailed in Tables 9 and 10). This matrix offers a visual comparison between the actual and predicted classifications, illuminating both the precision of the system and the nature of the errors. It was generated by contrasting the system's predictions with the established ground truth for each entry in the EXIST test dataset. Our focus is on understanding

the specific characteristics of sexism that make it challenging to accurately categorize sexist expressions. First, we focused on sexist messages that were misclassified as non-sexist. We identified 507 tweets, with 143 belonging to the ideological category, 118 to stereotypes, 102 to misogyny, 83 to sexual-violence, and 61 to objectification. The misclassification of ideological tweets can be attributed to some implicit and emerging sexism mechanisms such as a “flipped narrative” [4], where men are presented as victims of gender-based oppression or attacks on the feminist movement, sometimes as part of a political ideology (“*First it's a day without immigrants. Now we have a day without women. Why don't we have a day without liberals. We could really get some things done then.*”). Messages involving sexual violence and misogyny also pose challenges due to their complex patterns, such as reported situations or sarcasm (“*To hit women or shield them from bullets....life's choices are so difficult.*”). No significant patterns were observed for the remaining categories due to the variety of situations encountered. Some examples include implicit stereotypes (“*When we were at the university, we put a rule, 2 minutes we wait for the men, 5 for the women. I think most of us adopted that rule for any time. Today, no more than 2 minutes should we wait for someone*”), words that can be used in both sexist and non-sexist contexts, or explicit comments (“*Who got some fat titties I can rest my head on? it's been a long day*”).

Table 6 Percentage of sexist texts, per category and language, containing terms from the Hurltex lexicon

Language	Category	% Texts	Avg. tokens in Hurltex	Avg. number of words	Avg. characters
English	Ideological-inequality	51.03	1.5	38.1	235.4
	Misogyny-non-sexual-violence	62.67	2	26.8	154.5
	Objectification	68.75	1.7	27.4	158.8
	Sexual-violence	66.86	2	30.6	182.6
	Stereotyping-dominance	55.46	1.6	33.4	190.6
Spanish	Ideological-inequality	38.33	1.4	37.3	229.8
	Misogyny-non-sexual-violence	60.84	1.5	23.1	140.9
	Objectification	58.60	1.4	26.9	154.1
	Sexual-violence	53.17	1.6	28.8	169.3
	Stereotyping-dominance	42.88	1.3	30.3	177.9

Table 7 Predictions performed by two classifiers for the EXIST 2021 English dataset (only sexist messages)

Classifier	Predictions	% Texts
Davidson et al. [26]	Neither	71
	Hate	29
ToxiGen (Hartvigsen et al. [27])	Neither	49
	Hate	51

Second, we analyzed non-sexist tweets that were misclassified as sexist. We found 452 non-sexist messages that had been classified as ideological (79), misogyny (97), objectification (80), sexual-violence (91), or stereotypes (105). We observed that messages classified as ideological tended to use similar vocabulary to position themselves against an ideological movement, such as feminism, pornography, LGBT movements, abortion, or racism. Some of these messages discuss these topics without direct attacks or discrimination against women (“Where are all the white women at?”). For the misogyny and sexual violence categories, we found that most of the misclassified messages were extremely violent and aggressive, but they did not necessarily target women specifically. Instead, they sometimes targeted other communities or individuals regardless of their gender (“If you have a dick and balls you are a man. If you have a vagina and ovaries you are a woman. This is not hard, grab what you got between your legs and live with it. <https://t.co/CXdPrEcpgC>”). Regarding non-sexist messages that were misclassified as objectification or stereotypes, we find that the presence of terms referring to the physical attributes of women (e.g., “tits”, “bum”, etc) is usually interpreted as objectification or stereotyping, especially when paired with violent or hateful terms (“no, man, no... the gays don’t wanna see your tits in your penguin pyjamas”). This observation is interesting because it indicates that messages containing aggressive language or discussing controversial topics are more likely to be wrongly classified as sexist. This demonstrates that even advanced language models that are context-dependent and transformer-based are unable to completely eliminate all biases and accurately detect sexism.

Table 8 Predictions performed by two classifiers for the EXIST 2021 Spanish dataset (translated to English as a previous step for classification, only sexist messages)

Classifier	Predictions	% Texts
Davidson et al. [26]	Neither	79
	Hate	21
ToxiGen (Hartvigsen et al. [27])	Neither	59
	Hate	41

Notably, we encountered some instances of possible annotation mistakes for non-sexist messages. These errors may have occurred for various reasons. For example, we observed some cases where new forms of sexism, such as the flipped narrative, were not correctly identified, as seen in “@mitchellekaroro @eattherude_dude I hope that is aimed at some women because I asked a woman if she wanted sex after a few days I had known her and she accused me of sexual harrasmentIt’s a no win situation today”; additionally, for messages using insults or offensive terms without any context, it was sometimes not possible to know whether they were attacking a woman (“@thickiepops_art BE A WHORE!!”).

Finally, we analyzed errors in classifying sexist messages into the different categories defined in EXIST 2021. We found that 536 tweets were misclassified, with 89 in the ideological category, 162 in the misogyny category, 107 in the objectification category, 65 in the sexual violence category, and 104 in the stereotype category. Our analysis revealed that more subtle types of sexism, such as stereotypes or ideological messages, tended to be mistaken for each other due to their shared patterns (“I’d say a slightly better variant would be if the accusation comes at a politically inopportune time and without evidence. In the future they will use Republican women against Republican men. Like the Jews stabbing Trump in the back, women are inherently leftist.”). In fact, we found a significant number of messages that could be considered both stereotyping and ideological at the same time (“I am #antifeminist because I love to be treated like a lady by a gentleman.”). However, when stereotyping or ideological messages are expressed with hate, they are commonly classified as misogyny (“They want the ”framework for discussion” to agree with them hahahaha the sororas who are later more bitches than anyone with other women”). More aggressive subtypes of sexism such as misogyny and sexual violence are frequently mistaken for each other (“And as a whore ”as you say you are”, I would like to know why Andalusian did the colonoscopy you have with his fat cock. That happens to you, for fucking Andalusians.”). In addition, we noticed that some misogynistic texts tend to be classified as ideological when talking about politics (“I estimate that the amount of damage women do to society is somewhere in the range of trillions of dollars. The funding of various governmental women’s organizations are billions up millions.”). The objectification class is sometimes mistaken for the stereotype class when talking about women’s physical attributes (“make sure if your girlfriend is getting fat that you bring it up to her right away. women aren’t nearly as sensitive about these things as men think- and she’ll thank you for your candor. i’m SURE that she will understand”).

The examples provided above highlight the inherent subjectivity involved in interpreting sexist behaviors, which are

Table 9 Best run results for the task of sexism identification (task 1) in the EXIST 2021 competition per class

Team	Accuracy	F1	F1_non-sexist	F1_sexist
AI-UPV_1	0.7804	0.7802	0.7732	0.7872
SINAI_TL_1	0.7799	0.7797	0.7718	0.7875
IT_FHSTP_2	0.7754	0.7751	0.7682	0.7821
Multiaztertest_1	0.7740	0.7731	0.7586	0.7876
NLP_UNED_team_1	0.7719	0.7702	<u>0.7501</u>	0.7903
Free_1	0.7708	0.7707	0.7673	0.7741
GuillemGSubies_2	0.7683	0.7683	0.7668	<u>0.7697</u>
LHZ_1	0.7664	0.7660	0.7562	0.7759
Zk_1	0.7646	0.7644	0.7581	0.7708
Alclatos_1	0.7637	0.7635	0.7570	0.7700

influenced by external or cultural factors. This makes sexism detection and categorization a challenging task that requires practical knowledge. Furthermore, our findings suggest that categorizing sexist messages is a multilabel task, as texts often contain two or more types of sexist attitudes.

3.2.3 Examining consistency in characteristics of sexism messages across languages and its impact on automatic detection

In this subsection, we evaluate the performance of the systems presented by the participants for each of the two languages: English and Spanish. Through this analysis, we aim to identify the main differences in the expression of online sexism between the two languages and evaluate the feasibility of using cross-lingual techniques to address this problem. It is important to note that, given the mechanisms that were employed to gather the data and to avoid bias in the selection of authors when developing the EXIST dataset, we assume that the user profiles in both languages are comparable [4].

Tables 12 and 13 show the results of the top 10 systems for Task 1 (sexism identification) for English and Spanish, respectively. We observe that, taking into account both classes, English texts are more difficult to classify than Spanish texts (2.3% lower performance in terms of accuracy). However, the *sexist* class is predicted slightly better for English texts than in Spanish texts, while the *non-sexist* class is better predicted in Spanish.

Tables 14 and 15 display the results of Task 2 in terms of F1 score for each sexism category and language. The performance of the different teams varies significantly across some of the categories in both languages. In English, the most notable difference is observed in the *misogyny-non-sexual-violence* category, with a 12.2% variation between the best- and worst-performing teams. Similarly, in Spanish, there is an approximately 13.5% difference in the F1 score for the same category. This finding suggests that the performance

of the systems may depend on the specific category and language being evaluated and that different approaches may be needed to address sexism in different contexts.

It is worth noting that although the distribution of sexism subcategories differs between English and Spanish (see Table 2), the most and least challenging subcategories to detect are consistent across both languages. As discussed in Section 3.2.1, the English subset has a greater proportion of texts related to sexual violence (20% versus 9.9% in the training set) and contains more offensive messages than does the Spanish dataset. On the other hand, the Spanish training set has approximately 5% more tweets related to misogyny and ideology than the English training set. Despite these differences, the consistency in the difficulty of detecting certain subcategories across languages suggests that certain aspects of online sexism may be universal and that cross-lingual techniques may be effective in addressing them.

To investigate whether the performance differences among sexism categories can be attributed to differences in the consistency of the vocabulary used within each category, we computed the Jaccard similarity coefficient for the messages in each category separately for both English and Spanish. Table 16 presents the Jaccard similarity coefficients by category and language. Across all categories, Spanish messages exhibit greater vocabulary similarity, which may explain the better performance in task 1. In task 2, we observe notable differences in vocabulary homogeneity between the *misogyny-non-sexual-violence* and *sexual-violence* categories in English and Spanish, which might account for the lower performance of English texts. The most lexically homogeneous category in Spanish is *ideological-inequality*, which, together with its higher proportion in the Spanish training set, suggests the existence of a significant anti-feminist movement on Twitter in Spain. Interestingly, for English tweets, despite having similar levels of lexical heterogeneity in the *misogyny-non-sexual-violence* and *sexual-violence* categories, there is better performance on the latter category, likely due to the presence of more explicit

Table 10 Best run results for the task of sexism categorization (task 2) in the EXIST 2021 competition per class

Team	F1	F1_non-sexist	F1_ideological-inequality	F1_stereotyping-dominance	F1_objectification	F1_sexual-violence	F1_misogyny-non-sexual-violence
AI-UPV_1	0.5787	0.7732	0.6435	0.5065	0.4859	0.5957	0.4674
LHZ_1	0.5705	0.7633	0.6511	0.5058	0.46	0.5974	0.4454
SINAI_TL_1	0.5667	0.7691	0.6356	0.5026	0.4688	0.5595	0.4646
QMUL-SDS_1	0.5594	0.7601	0.6578	0.5032	0.4556	0.5682	0.4113
IT_FHSTP_2	0.5589	0.7682	0.6426	0.4685	0.4539	0.5767	0.4434
Alclatos_1	0.5578	0.757	0.619	0.4896	0.4779	<u>0.5263</u>	0.477
IREL_hatespeech_group_3	0.5555	0.7538	0.64	0.4696	<u>0.4517</u>	0.5484	0.4689
Zk_1	0.552	0.7662	0.6533	0.4892	0.442	0.538	0.4235
NLP_UNED_team_3	0.55	0.7419	<u>0.6164</u>	<u>0.4667</u>	0.4968	0.5414	0.4421
Recognai_1	0.55	<u>0.7136</u>	0.6374	0.4752	0.4654	0.6013	<u>0.407</u>

The bold entries indicate the best results in relevant cases discussed in the manuscript

Table 11 Average Jaccard coefficient for the vocabulary within the different categories of sexism

Category	Jaccard similarity
All categories	0.047
Stereotyping	0.055
Misogyny	0.040
Objectification	0.047
Ideological	0.064
Sexual-violence	0.047

The bold entries indicate the best results in relevant cases discussed in the manuscript

sexual and hateful language, as noted in the experiments in Section 3.2.1.

3.2.4 Investigating the influence of social network platforms on sexism detection

The objective of this subsection is to assess whether algorithms trained on sexist expressions from Twitter can be used to detect sexism on other social networks, such as Gab, or whether the characteristics of the network affect how sexism is expressed, requiring a specific learning approach. Notably, the user profiles of the two networks are likely to be different.

While male young adults are the predominant demographic in both networks, Gab is known to have a far-right userbase [29], attracting users from other social platforms that implement censorship mechanisms [30]. Consequently, sexism on Gab is expected to be expressed in a more explicit, open, or even violent manner. Conversely, hateful and offensive forms of sexism are more likely to be censored on Twitter.

Table 17 presents the average performance results of the top 10 participants in the sexism detection task when trained on tweets and tested on both tweets and gabs. Interestingly, the results show that the systems perform slightly better on gabs than on tweets, indicating that learning to detect sexism on Twitter can also be useful in identifying sexism in other social networks.

Table 18 presents the comparison between the results obtained by the top 10 participants in the categorization task for each sexism category and social network. While similar results are observed for the *ideological-inequality* category in both networks, significant differences exist among the other categories. Notably, the performance difference between Gab and Twitter is more pronounced in the *misogyny-non-sexual-violence* class, which is poorly predicted in Gab. Our hypothesis, as discussed in Section 3.2.1, is that misogyny on Twitter is often expressed using nonviolent and nonhateful

Fig. 1 Confusion matrix for the best system in EXIST 2021 for task 2 (y-axis True label)

ideological-inequality	389	26	143	10	18	35
misogyny-non-sexual-violence	46	208	102	27	47	42
non-sexist	79	97	1,635	80	91	105
objectification	15	28	61	156	29	35
sexual-violence	14	26	83	19	252	6
stereotyping-dominance	45	33	118	26	9	233
	ideological-inequality	misogyny-non-sexual-violence	non-sexist	objectification	sexual-violence	stereotyping-dominance

Table 12 Best run results for the task of sexism identification (task 1) in the EXIST 2021 competition, when only the English texts are considered

Team run	Accuracy	F1	F1 _{non-sexist}	F1 _{sexist}
SINAI_TL_3	0.7771	0.7746	0.7510	0.7983
Multiaztertest_1	0.7717	0.7690	0.7441	0.7939
AI-UPV_1	0.7667	0.7657	0.7500	0.7811
Free_1	0.7667	0.7661	0.7544	0.7779
Zk_1	0.7663	0.7650	0.7480	0.7820
LHZ_1	0.7658	0.7633	0.7387	0.7878
NLP_UNED_team_1	0.7600	0.7571	<u>0.7269</u>	0.7872
GuillemGSubies_1	0.7600	0.7598	0.7484	0.7712
AIT_FHSTP_2	0.7595	0.7583	0.7413	0.7752
IREL_hatespeech_group_3	0.7576	0.7566	0.7404	<u>0.7728</u>

The bold entries indicate the best results in relevant cases discussed in the manuscript

language, while on Gab, it tends to be expressed with hostility and hatred. This pattern also holds for the *objectification* and *stereotyping-dominance* categories. On the other hand, since sexual violence is often expressed using crude and obscene language, this category is well predicted in Gab. This suggests that while algorithms trained on Twitter data can be used to detect sexism in other networks, the way sexism is expressed may differ depending on the platform and may require specific adaptations in the learning process.

3.2.5 Determining the most suitable algorithms, techniques, and resources for detecting sexist expressions in social networks

In this subsection, we analyze the methods and models used by the top 10 teams during EXIST 2021 to detect sexist content in social networks, in order to guide the development of new classification systems for future investigations [31]. Specifically, we compare the participating systems based on the following dimensions: type of transformer, transformer language, data augmentation, and classification framework. The “type of transformer” can take two values, “BERT” or “RoBERTa”. The “transformer language” dimension can have two values, “Monolingual” or “Multilingual”. The “data

augmentation” dimension has five different values: “Translation”, indicating whether participants augmented data by translating the dataset; “Labeled data” or “Unlabeled data”, indicating whether participants used external datasets to fine-tune or pre-train their transformers; “External lexicon”, indicating the use of any lexical resource; and “Multitask learning with other data”, indicating whether participants used a multitask learning approach. Finally, the “classification framework” dimension has four values: “Ensemble”, indicating the use of multiple transformer models simultaneously; “Stylometrics”, indicating the use of style features such as sentence length; “Extra classifier at the output”, indicating whether participants used an additional classifier at the output (other than linear layers); and “Cascade approach”, indicating that participants removed the non-sexist class from task 2 so that the models only predicted the type of sexism and then used task 1 to classify non-sexist tweets.

Table 19 presents the analysis for the first task of sexism detection. Consistent with recent trends in NLP, transformer-based models outperformed other techniques, with all the top 10 teams utilizing some form of transformer-based system for both tasks. Therefore, we focus our analysis on the type of transformer used.

Table 13 Best run results for the task of sexism identification (task 1) in the EXIST 2021 competition, when only the Spanish texts are considered

Team run	Accuracy	F1	F1 _{non-sexist}	F1 _{sexist}
AI-UPV_1	0.7944	0.7944	0.7950	0.7938
AIT_FHSTP_2	0.7916	0.7916	0.7935	0.7897
TL_1	0.7907	0.7905	0.7965	0.7845
NLP_UNED_team_1	0.7833	0.7827	0.7719	0.7936
Alclatos_1	0.7791	0.7791	0.7800	0.7782
GuillemGSubies_2	0.7763	0.7761	0.7840	0.7681
Multiaztertest_1	0.7763	0.7763	0.7720	0.7805
Free_1	0.775	0.7748	0.7796	0.7701
S_exist_1	0.7703	0.7699	0.7795	<u>0.7603</u>
LHZ_1	0.7671	0.7670	<u>0.7718</u>	0.7621

The bold entries indicate the best results in relevant cases discussed in the manuscript

Table 14 Best run results for the task of sexism categorization (task 2) in the EXIST 2021 competition, when only the English texts are considered

Team run	F1	F1_non-sexist	F1_ideological-inequality	F1_misogyny-non-sexual-violence	F1_objectification	F1_sexual-violence	F1_stereotyping-dominance
LHZ_1	0.5604	0.747	0.6386	0.4049	0.4968	0.5619	0.513
IREL_hatespeech_group_3	0.5531	0.7453	0.6139	0.4716	0.461	0.5686	0.458
AI-UPV_1	0.5507	0.7503	0.6126	0.4062	0.4833	0.5678	0.48393
Zk_1	0.5432	0.7561	0.6171	0.3915	0.4594	0.5425	0.4924
AIT_FHSTP_2	0.5418	0.7413	0.6028	0.4126	0.4726	0.5474	0.4745
SINAI_TL_3	0.5374	0.749	0.6212	0.4022	0.4258	0.5669	0.4595
NLP_UNED_team_1	0.5371	0.7423	0.5824	0.4142	0.4758	0.5137	0.494
QMUL-SDS_1	0.5351	0.7429	0.6267	<u>0.3492</u>	0.4615	0.5341	0.4962
Recognai_1	0.5251	<u>0.7108</u>	0.6096	0.4098	<u>0.4137</u>	0.5502	<u>0.4567</u>
Alclatos_1	0.5245	0.733	<u>0.5728</u>	0.399	0.4761	<u>0.5011</u>	0.4647

The bold entries indicate the best results in relevant cases discussed in the manuscript

Table 15 Best run results for the task of sexism categorization (task 2) in the EXIST 2021 competition, when only the Spanish texts are considered

Team run	F1	F1_non-sexist	F1_ideological-inequality	F1_misogyny-non-sexual-violence	F1_objectification	F1_sexual-violence	F1_stereotyping-dominance
AI-UPV_1	0.6073	0.795	0.677	0.5138	0.4887	0.6321	0.537
SINAI_TL_2	0.6013	0.7858	<u>0.6503</u>	0.5405	0.4899	0.5907	0.5508
Alclatos_1	0.5921	0.78	0.6718	0.5383	0.4795	0.5578	0.5252
QMUL-SDS_1	0.5842	0.7764	0.6923	0.4581	0.4493	0.6172	0.5123
LHZ_1	0.5804	0.7782	0.6655	0.4787	<u>0.4192</u>	0.6449	0.4962
AIT_FHSTP_2	0.5761	0.7935	0.686	0.468	0.4338	0.6149	<u>0.4604</u>
NLP_UNED_team_3	0.5759	0.7606	0.6507	0.4915	0.523	0.5537	0.4761
Codec_1	0.5759	0.7878	0.6544	0.493	0.4832	0.5483	0.4885
Recognai_1	0.5749	<u>0.7167</u>	0.6624	<u>0.4046</u>	0.5027	0.6646	0.4986
GuillemGSubies_2	0.5646	0.784	0.6655	0.47294	0.4554	<u>0.5243</u>	0.4854

The bold entries indicate the best results in relevant cases discussed in the manuscript

Among the top 5 teams, two exclusively utilized BERT, two employed only RoBERTa, and the winning team, *AI-UPV* [32], used both in an ensemble. Their solution involved a monolingual setup with BERT for English, BETO [33] for Spanish, and mBERT for both Spanish and English. They also implemented individual models with translation for both English and Spanish texts. RoBERTa was consistently utilized by the top 5 teams in a multilingual setting, whereas BERT was employed exclusively in a monolingual setting. Among the remaining teams, four used both BERT and RoBERTa, three used a multilingual configuration and one used BERT monolingually. These results suggest that using both transformers simultaneously is not effective unless they are combined through ensemble techniques. Surprisingly, the multilingual RoBERTa outperformed its monolingual counterpart, supporting its effectiveness in a single-model approach for detecting sexism on social networks.

Additionally, we observe that monolingual BERT models perform better when paired with data augmentation techniques, as demonstrated by the second-place results achieved by the *SINAI_TL_1* team. This suggests that monolingual versions benefit by using additional data to enhance their performance. In particular, this team employed an alternative multitask learning approach, incorporating various auxiliary tasks and employing BETO for Spanish and BERT for English as base models. Their most successful runs involved using polarity classification as an auxiliary task and training a shared model with the InterTASS dataset [34]. However, monolingual BERT did not perform as well as expected when combined only with back translation (*GuillemGSubies_2* team).

Data augmentation techniques seemed to improve the results because four out of the top five systems utilized some form of data augmentation. Notably, back translation and the use of labeled data proved to be effective, as they were used by the top three teams. In contrast, only one of the five worst-performing teams utilized data augmentation. Additionally, it is worth noting that multilingual transformers appear to perform better than monolingual ones when not using data augmentation techniques, as seen in the case of the *NLP_UNED_team*.

In terms of the classification framework dimension, it is worth noting that the two teams that employed a cascade approach, where messages predicted as sexist in task 1 were considered in the sexist categorization for task 2, performed among the top 5. The addition of stylometric features did not enable systems to outperform the top three teams. Interestingly, ensemble techniques appear to be the most effective option when used in combination with data augmentation techniques.

Finally, it is worth mentioning that some participants who did not rank in the top 10 utilized traditional machine learning techniques such as support vector machines (SVMs), random

Table 16 Average Jaccard distance coefficient between pairs of tweets for each category and language

Language	Category	Jaccard similarity
English	All categories	0.0390
	stereotyping	0.0480
	misogyny	0.0370
	objectification	0.0472
	ideological	0.0537
	sexual-violence	0.0370
Spanish	All categories	0.0540
	stereotyping	0.0620
	misogyny	0.0440
	objectification	0.0460
	ideological	0.0760
	sexual-violence	0.0590

The bold entries indicate the best results in relevant cases discussed in the manuscript

forests (RFs), and logistic regression (LR), as well as other deep learning techniques such as long short-term memory networks (LSTMs). However, despite incorporating additional features based on external resources, these methods did not yield satisfactory performance. Similarly, two teams unsuccessfully attempted to utilize external lexicons.

To summarize, our analysis reveals that data augmentation techniques are effective in improving the performance of transformer models, particularly for monolingual transformers. This could be attributed to the fact that sexism is language-specific, and thus, a monolingual model performs better when trained on a sufficient amount of data. Additionally, our findings suggest that the ensemble technique benefits the most from data augmentation, which is consistent with previous research on hate speech detection [35].

Table 20 presents the analysis for the second task, sexism categorization. The conclusions drawn from this analysis are similar to those of the first task. However, we observe that some approaches, such as the one used by the *LHZ* team who placed second, are based on monolingual transformers. Their solution utilized a different transformer model for each language: DeBERTa [36] for English and XLM-R for Spanish. An LSTM network was then applied to each representation to obtain the final predictions. This improvement

Table 17 Average results for the task of sexism identification (task 1) in the EXIST 2021 competition, when only tweets and gab posts are considered for testing, respectively

Platform	Accuracy	F1	F1_non-sexist	F1_sexist
Twitter	0.7708	0.7704	0.7629	0.7780
Gab	0.7769	0.7759	0.7617	0.7901

The bold entries indicate the best results in relevant cases discussed in the manuscript

Table 18 Average results for the task of sexism categorization (task 2) in the EXIST 2021 competition, when only tweets and gab posts are considered for testing, respectively

Platform	F1	F1_non-sexist	F1_ideological-inequality	F1_misogyny-non-sexual-violence	F1_objectification	F1_sexual-violence	F1_stereotyping-dominance
Twitter	0.5696	0.7581	0.6336	0.4772	0.4849	0.5553	0.5087
Gab	0.5290	0.7400	0.6543	0.3676	0.4038	0.6031	0.4052

The bold entries indicate the best results in relevant cases discussed in the manuscript

Table 19 Classification techniques used by the top 10 teams in the detection task (task 1) of the EXIST 2021 competition

Team	Transformer based on		Transformer language		Data augmentation		Classification framework		Extra clas- sifier	Cascade
	BERT	RoBERTa	Monolingual	Multilingual	Translation	Labeled data	External Lexicon	Multitask learning		
AI-UPV_1	✓	✓	✓	✓	✓				✓	✓
SINAI_TL_1	✓		✓			✓		✓		
AIT_FHSTP_2		✓		✓	✓	✓				
Multiaztertest_1	✓		✓				✓			
NLP_UNED_team_3		✓		✓				✓		✓
Free_1	✓	✓	✓							
GuillemGSubies_2	✓		✓		✓					
LHZ_1	✓	✓	✓						✓	
Zk_1	✓	✓	✓	✓					✓	
Alclatos_1	✓	✓	✓						✓	

Table 20 Classification techniques used by the top 10 teams in the categorization task (task 2) of the EXIST 2021 competition

Team	Transformer based on		Transformer language		Data augmentation		Classification framework						
	BERT	RoBERTa	Monolingual	Multilingual	Translation	Labeled data	Unlabeled data	External Lexicon	Multitask learning	Ensemble	Stylometrics	Extra classifier	Cascade
AI-UPV_1	✓	✓	✓	✓	✓				✓				✓
LHZ_1	✓	✓	✓									✓	
SINAI_TL_1	✓		✓			✓			✓				
QMUL-SDS_1		✓		✓				✓				✓	
AIT_FHSTP_2		✓		✓	✓								
Alclatos_1	✓	✓	✓										
IREL_hatespeech_group_3		✓	✓		✓		✓	✓		✓	✓	✓	
Zk_1	✓	✓	✓	✓						✓			
NLP_UNED_team_3		✓		✓					✓				✓
Recognai_1	✓	✓	✓										✓

can likely be attributed to the use of an additional classifier at the transformer output (as in the case of the LSTM layer for *LHZ*), which was employed by two out of the top five teams. Interestingly, the *IREL_hatespeech_group* attempted to use unlabeled data augmentation techniques but did not achieve successful results.

3.3 Designing a system for detecting and categorizing online sexism

In this section, a novel approach for detecting sexism is presented, which combines all the findings discussed in this article into a single, straightforward two-step method. In the first step, we pre-train two monolingual transformer architectures using unlabeled tweets; in the second step, we fine-tune these architectures with the EXIST 2021 labeled dataset.

We have developed two hypotheses based on our findings. First, we propose that since sexism appears to be language-specific (see Section 3.2.3), monolingual models may perform better when trained on a similar volume of data as multilingual models. Second, we observe that differences in performance across subtypes of sexism may be related to variations in vocabulary, as discussed in Section 3.2.2. To address this issue, we aim to explore data augmentation techniques to mitigate vocabulary differences by increasing the amount of data used in the pre-training stage, particularly for minority classes or those with high heterogeneity such as *misogyny-non-sexual-violence*. These hypotheses are consistent with the findings of the shared task outlined in Section 3.2.5, which suggests that combining monolingual models with data augmentation techniques can improve the performance of sexism detection systems.

The architectural design employed for both tasks in EXIST 2021 is shown in Fig. 2. Our strategy incorporates a two-step approach. Initially, we pre-trained two separate models: MarIA [37] for Spanish and RoBERTa [38] for English. These models served as the foundation for our archi-

tecture. To accumulate data for this phase, we followed the methodologies outlined by the EXIST organizers [4] and amassed unlabeled Twitter data, subsequently preprocessing it by standardizing and replacing all URLs and usernames with “twurl” and “twuser”, respectively. This effort culminated in a dataset of approximately 6 million tweets, equally split between the two languages. During this phase, our central objective was to increase the models’ ability to generalize, especially given the limited amount of labeled data from the competition. The training utilized a masked language modeling approach, where 15% of the tokens were chosen at random to be masked, with predictions being made using cross-entropy loss.

After the pre-training, an integration layer was added to the models. This was not a simple fusion of the two pre-trained models but a more intricate combination. This layer incorporated a dropout mechanism with a 0.1 probability and a subsequent fully connected layer, aiming to blend insights from both linguistic models into a unified output. The models were then fine-tuned using the EXIST 2021 dataset in both English and Spanish. This approach combined the strengths of the two powerful models and introduced a mechanism to capture a nuanced understanding of sexism in two distinct languages.

4 Evaluation and results

In this section, we analyze and discuss the results obtained from the experiments. We also provide details on our evaluation methods and error analysis to provide a comprehensive understanding of our findings.

4.1 Experimental setup

During pre-training, the models were trained on two NVIDIA RTX 3090 GPUs, using the Adam [39] optimization algo-

Fig. 2 Classification system for EXIST tasks 1 and 2

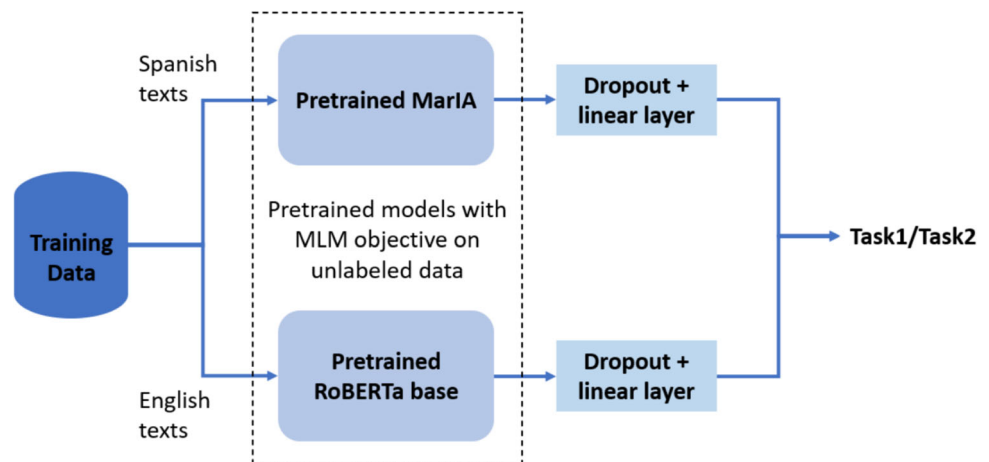


Table 21 Results of EXIST 2021 task 1, baseline and comparison of techniques

System	Accuracy	F1	Recall	Precision
SVM	0.6845	0.6832	0.6888	0.6943
XLM-R	0.761 (↑ 7.6%)	0.760	0.761	0.760
Our system	0.7919(↑ 3%)	0.7918	0.7928	0.7923

The bold entries indicate the best results in relevant cases discussed in the manuscript

rithm with a learning rate of $1e-4$ and a linear weight decay of 0.01. The training batches consisted of 32 samples spanning ten epochs.

For fine-tuning, training was performed on an NVIDIA RTX 3090 GPU, using the Adam optimizer [39] with an adjusted learning rate of $2e-5$ while keeping the weight decay at 0.01. This phase used batch sizes of 16 and ran for four epochs. All modeling and experiments utilized the PyTorch library [40], supplemented by the Transformers library from HuggingFace [41].

4.2 Results

Tables 21 and 22 present the results of our experiments on EXIST 2021 tasks 1 and 2, comparing our proposed system to two different baselines: a traditional machine learning method based on SVM with TF-IDF features and a transformer-based baseline using an XLM-R multilingual transformer. We evaluate the effectiveness of our approach using four metrics (computed as the average of all the classes): accuracy, precision, recall, and F1 score. To facilitate comparison with the competition, we report the accuracy for Task 1 and F1 score for Task 2, as these were the evaluation metrics used for ranking purposes.

For task 1, in line with recent advances in NLP, the XLM-R transformer model showed significant improvement over the SVM model, highlighting the benefit of contextual embeddings from transformer-based models for text classification tasks. Our proposed system achieved the highest scores across all the metrics. With an accuracy of 0.7919, our system outperformed the XLM-R transformer baseline by 3% and the SVM baseline by more than 10%.

The results of Task 2, which is more complex, show a more significant improvement of the proposed system. This aligns with our findings, which show that applying specific systems

to each language can better detect the nuances of each language. This explains the significant differences between our system and both baselines; there is an 11.64% improvement over the XLM-R and an approximately 19% improvement over the SVM.

4.3 Discussion

In this subsection, we extend the discussion to compare the robustness and efficiency of our system. In particular, we compare the performance of our system with that of the best solution from the EXIST competition. Table 23 presents the results of our proposed system on task 1 and task 2 compared to those of the best-performing team in the EXIST 2021 competition. As shown in task 1, our system achieves a 1.15% increase in accuracy, the metric used for ranking in the competition. This improvement is remarkable given that the difference between the top 10 runs in the competition is approximately 1.39% [4]. In task 2, our system achieves a 0.7% improvement in the F1 score, which is the metric used for ranking in the competition, achieving the best result among the shared task participants. These results support our hypotheses: data augmentation improves the performance of transformer-based models, particularly when more data is available, and sexism detection should be handled separately for each language. It is also noteworthy that our system is more computationally efficient because it only uses two models, in contrast to AI-UPV, which employs an ensemble of five models.

To increase the robustness of our results, we evaluated our system on the 1,052 new tweets that were used as the test set in EXIST 2022 after training it on the entire EXIST 2021 dataset. Table 24 shows the results of this evaluation and the comparison to the best team in that competition, AvaCaonda [42], which proposed, similar to the best solution in EXIST 2021, proposed an ensemble of various transformer models, including BERTweet-large [43], RoBERTa, and DeBERTa for English, and BETO, BERTIN [44], MarIA, and RoBERTuito [45] for Spanish. As the table shows, our system outperforms the best one in all metrics. Similar to EXIST 2021, the top-performing solution consists of large ensembles of multiple transformer models, presenting a challenging baseline due to the number of models employed and their substantial predictive capacity. However, our system

Table 22 Results of EXIST 2021 task 2, baseline and comparison of techniques

System	F1	Accuracy	Recall	Precision
SVM	0.3950	0.5222	0.3772	0.3950
XLM-R	0.47 (↑ 8%)	0.66	0.5	0.45
Our system	0.5864(↑ 11.64%)	0.6708	0.5691	0.6102

The bold entries indicate the best results in relevant cases discussed in the manuscript

Table 23 Results for the sexism identification and categorization tasks and comparison with the best performance system in EXIST 2021

Task	System	Accuracy	F1	Recall	Precision
Task 1	Best System EXIST 2021 [32]	0.78	0.78	0.78	0.78
	Our system	0.7919	0.7918	0.7928	0.7923
Task 2	Best System EXIST 2021 [32]	0.6557	0.578	0.577	0.581
	Our system	0.6708	0.5864	0.5691	0.6102

The bold entries indicate the best results in relevant cases discussed in the manuscript

Table 24 Results for the sexism identification and categorization tasks and comparison with the best performance system in EXIST 2022

Task	System	Accuracy	F1	Recall	Precision
Task 1	Best System EXIST 2022 [42]	0.799	0.797	0.797	0.798
	Our system	0.804	0.8015	0.801	0.8021
Task 2	Best System EXIST 2022 [42]	0.7013	0.5106	0.5112	0.51
	Our system	0.7108	0.513	0.5154	0.522

The bold entries indicate the best results in relevant cases discussed in the manuscript

Table 25 Examples labeled by the model and the best EXIST 2021 system

N	Text	Real	Predicted by AI-UPV system	Predicted by our system
(1)	Man left with criminal record after calling wife a slapper, it's illegal to be unfriendly to your wife after she's cheated on you	Misogyny-non-sexual-violence	Sexual-violence	Misogyny-non-sexual-violence
(2)	And they learn that they can't trust anything women actually say. If they express appreciation for a woman, they discover that this makes them seem weak and unattractive. Do you see how the big man man that you're complaining about is *men reacting to the way women treat them?*	Misogyny-non-sexual-violence	Stereotyping-dominance	Misogyny-non-sexual-violence
(3)	@user No.... there is no reason to, from a Males stance. Every law is stacked against men #MGTOW	Ideological-inequality	Non-sexist	Ideological-inequality
(4)	If you have a dick and balls you are a man.If you have a vagina and ovaries you are a woman. This is not hard, grab what you got between your legs and live with it.	Non-sexist	Sexual-violence	Non-sexist

Table 26 Examples labeled by the model

N	Text	Real	Predicted
(1)	@user ass hell! I would always want to sodomize you so impossible to be just your friend	sexual-violence	non-sexist
(2)	Yesterday, here in MA.A 45 year old woman driving the wrong way on 495 in Westford killed a 35 year old man at 2 AM. This morning the police released the name of the man killed. Still won't release the woman's name. I'm going to bet she is here illegally. Any takers?	ideological-inequality	non-sexist
(3)	Absolutely not. You can't blame all women for the actions of some others.	non-sexist	ideological-inequality
(4)	Why are you so obsessed with men having sex with other men? Seriously, your response to everything involves accusations of homosexuality or cocksucking. You are clearly a repressed faggot	non-sexist	sexual-violence

outperforms it by employing only two transformer models, resulting in a much more efficient solution.

To provide deeper insights, we conducted two manual inspections. First, we compared some of our correct predictions with the errors made by the best EXIST 2021 system. Second, we analyzed the errors that were transferred and retained in our system. Table 25 presents a series of texts that were incorrectly identified by the top-performing model in the EXIST 2021 competition that our proposed system was able to identify correctly. The table reveals that our model accurately categorized several complex examples.

Table 26 presents some examples of misclassifications made by the proposed system, which we manually examined. In example (1), our classifier failed to detect sexism, possibly due to the tweet's length and the absence of explicit sexist language. Example (2) is a challenging case of indirect sexism, where the narrative is flipped, making it difficult for the model to identify. Examples (3) and (4) are non-sexist messages that were misclassified as sexist, likely because of the presence of terms such as "all women" or "having sex", which may have biased the model toward the sexist category. Increasing the amount of data in the training stage could help the system reduce these types of errors.

5 Conclusions and future work

This paper presents a detailed analysis of sexism, exploring its characteristics and varied forms across multiple languages on social networks using computational approaches. By utilizing the EXIST 2021 competition's resources for online sexism detection, our first research question aims to validate the suitability of the EXIST dataset for our study and for detecting sexism in a broader sense. Our experiments, enhanced by external tools such as the HurtLex lexicon, confirm the dataset's effectiveness in capturing both explicit and nuanced sexism.

In our second research question, we examined different methods for sexism detection. The results show that data augmentation significantly enhances detection, especially with monolingual transformers, due to the language-specific nature of sexism and its varied expression across cultures.

Our study explored the linguistic aspects and nuances of sexism through two research questions. First, we examined the types of sexism depending on the facet of women being attacked (Research question 3), finding that while misogyny can manifest subtly and be challenging to detect, instances of sexual violence are overt and easier to identify. Second, we compared sexist expressions in English and Spanish (Research question 4), noting more sexual and hateful messages in English and a stronger hostility toward feminism and women's rights in Spanish. Despite the language differences, the most difficult categories to predict remained consistent.

Furthermore, an error analysis of EXIST 2021's top system underscored the difficulty of sexism detection, pointing to the subjectivity involved in interpreting sexist content. Overall, this study sheds light on the complexities of identifying sexism and highlights the need for nuanced approaches to address it.

Additionally, we investigated whether the sexism observed in conventional social networks such as Twitter can be utilized to identify sexism on the uncensored platform Gab (Research question 5). Our findings confirm that sexist language on Twitter can be a useful indicator of sexism on Gab, where such speech tends to be more aggressive and obvious. Furthermore, we developed a monolingual transformer-based system that leverages unlabeled in-domain data during its pre-training phase. This system outperforms the current state-of-the-art methods in the field by using only two transformer models (Research question 6).

Notably, our work has several limitations. First, the EXIST dataset does not include demographic information (such as gender, age, and ethnicity) regarding either the authors of the potentially sexist messages or the annotators of the dataset. This omission is largely due to the use of crowdsourcing for labeling and the preservation of anonymity within social networks. Nonetheless, the development of EXIST 2021 took into account various sources of potential bias to ensure the representativeness of the dataset [4].

Second, to adequately address concerns regarding the application of our findings in real-world scenarios, it is crucial to recognize the ethical implications of deploying automated sexism detection systems. Protecting personal privacy, ensuring accurate identification, and minimizing false positives are of greatest importance. In this study, our analysis focuses on textual content, without utilizing any personal information about the message creators or data obtained directly from them. It is important to clarify that the system's purpose is to identify texts containing or expressing sexism. However, it does not aim to classify or make judgments about specific individuals or accounts based on this analysis. Future research should continue to examine ethical considerations and explore methodologies that enhance the reliability of sexism detection technologies. This includes refining categorization for specific contexts, expanding language and platform coverage to achieve a more comprehensive understanding of sexist text, and incorporating contextual information to improve detection. In addressing the issue of false positives and misidentification, we are considering the implementation of a traffic light system. The system would be designed to utilize the classifier's probability scores to notify the author, recipient, or platform of the potential presence of sexist content in a message. The urgency of the alert would vary depending on the severity of the sexism category, with misogyny and sexual violence being treated as significantly more serious than other subtypes of sexism.

Finally, in future work we plan to address the sexism detection and classification problem from the perspective of *learning with disagreements* perspective. We recognize that perceptions of sexism are inherently subjective, making bias and fairness significant and unavoidable issues. To more effectively address this limitation, the EXIST 2023 competition was hosted as part of the CLEF 2023 conference [46] and proposed a new dataset using the learning-from-disagreements paradigm to consider different individual perspectives and points of view. Even though this challenge has shown that exploiting disagreement may improve the detection and classification of sexism, significant differences in sexism perception across genders and ages have not been found. In future work, we plan to extend the present study to this new paradigm, which will require new methods and analyses, as well as to analyze the effect of other sociodemographic characteristics such as educational background, religion and political orientation.

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Author Contributions All authors contributed equally to this work.

Data Availability Data will be made available on request.

Declarations

Competing Interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical and Informed Consent for Data Used The data utilized in this study was developed by the authors specifically for research purposes within the context of the EXIST competition [4]. Given its original creation and management by the authors, there are no concerns related to external data collection or participant consent. All necessary ethical considerations, including ensuring the anonymity and confidentiality of all participants or contributors, were strictly adhered to during data collection and processing.

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